

7.15 release Stack 01 audit — local candidate

The local source review and preflight are complete. Copilot's first authorized review of head cec3cac2487356e0ffa556d201e5eab14f31e054 found five issues; its second authorized review of 23b3c16b1b43e50ff04d8c5e4a77ee37c4a8e393 found six more. Remediation is in progress; a third review requires owner approval. Stack 00b predecessor acceptance and physical release checks remain pending.

Second Copilot review, 2026-09-23

The owner authorized exactly one additional Copilot request. GitHub recorded it at 07:19:00 UTC. Review 5288086285 assessed the exact remediation head 23b3c16b1b43e50ff04d8c5e4a77ee37c4a8e393 and recommended six new changes. The earlier RedPallas thread is still listed as open in the overview, but it is a finding on the old head, not a new inline comment in this review. No third request has been authorized or sent.

New finding	Audit response
Compose build failure can be ignored	CI now captures BUILD_RC, still extracts available reports, and requires a successful build along with both test containers. A stale local image cannot make the gate green after a failed build.
Newer-storage refusal can report successful settings changes	All persistence-entry guards in fsm.c now reject storage_isFirmwareTooOld() as well as the Bitcoin-only lock before modifying the RAM shadow.
Ripple memo support is unreachable	The unconditional early rejection in fsm_msgRippleSignTx() is removed. The existing confirmation and serializer paths handle a nonempty memo.
Zcash privacy signing is not wired	The regular image now includes fsm_msg_zcash.h, registers its six inbound and six outbound message types, and aborts its signing state with other workflows. A native test checks the map in the regular image and absence in Bitcoin-only. This also makes the earlier RedPallas production call part of the compiled firmware.
Reset can report success after storage refuses the write	setup_commit() now checks both storage downgrade locks before applying staged settings and seed, aborts the ceremony, and sends a failure.
Duplicate metadata source	The second signed_metadata.c entry is removed from the CMake target.

The second-review retrospective is that the first remediation addressed the five inline symptoms but relied on the existing green CI graph as evidence of the surrounding product. That graph did not prove the Zcash handler was linked, and its Compose step could have reused a stale image after a build failure. The source audit also missed a second storage refusal state and a contradictory early Ripple guard. The new checks cover the build gate and actual production message map, then require both build variants to compile. The original 140-row message-map parity statement below describes the original adjacent commit; the new Zcash mapping deliberately changes the audit remediation head's map. The final remediation head and hosted CI result will be recorded in PR #831's live description and the main-worktree claim without altering this reviewed source receipt solely to update a SHA.

Copilot review and audit retrospective, 2026-09-23

GitHub recorded one review_requested event at 06:29:19 UTC and review 5287704828 at 06:41:50 UTC. Its commit_id matches the frozen head cec3cac2487356e0ffa556d201e5eab14f31e054. The review recommended changes: two high, two medium, one low. No second request had been sent at this first-review checkpoint.

Finding	Evidence and remediation
Dice entropy through DebugLink	reset_get_int_entropy() returned the live int_entropy bytes while MIXED/ONLY reset was active, including the final dice-derived seed. The dice mode is now set before the draw and DebugLink omits this field throughout either dice ceremony. The intentionally displayed 24 MIXED entropy words remain on the device.

Finding	Evidence and remediation
RedPallas production call	<code>fsm_msg_zcash.h</code> passed four arguments to a five-argument signing API. The caller now obtains a health-checked 80-byte nonce transcript, passes it to RedPallas, and clears it on all paths. At this first-review checkpoint, the header was not included by <code>fsm.c</code> and its handlers had no message-map entries, so existing CI could not compile this production call. The second-review remediation above wires that path.
Malformed tiny frame	<code>msg_read_tiny()</code> silently returned for non-64-byte frames, leaving blocked handlers polling. It now sends the one-shot failure and unwinds.
Rejection latch lifetime	<code>tiny_handler_rejected</code> survived into a later tiny poll. Each independent poll now clears it before reading new input. A native UDP regression exercises malformed-frame cancellation followed by a valid acknowledgement.
Incorrect audit invariant	The Bitcoin-only dice SOP claimed that production never renders the device draw. MIXED mode intentionally shows it as 24 words before rolls. The invariant now states that behavior and distinguishes on-device display from raw DebugLink disclosure.

Why the internal audit and green CI missed these: review focused on successful transport and dice derivation paths, not malformed tiny-frame recovery and the state of the next poll. The dice review relied on an SOP invariant that was already false, and did not trace DebugLink while reset was active. Zcash unit tests compiled the RedPallas API directly, while the production handler header is dormant, so neither those tests nor firmware CI type-checked its caller. The next audit pass must trace each confidential value through debug responses, exercise failure then recovery across handler boundaries, and verify every claimed production handler is both compiled and mapped in each build variant.

The later sections retain the frozen intake and pre-review evidence. Their older checkpoint wording is superseded by this dated review record. The reviewed head's [exact-head CI run 35823135480](#) passed before these fixes; it is not evidence for the remediation head.

Identity and scope

- Owner: Codex /root, `keepkey-vault-v11-agent-2`; main-worktree claim: `docs/release/audit-claims/715-01/OWNER.md`.
- Main-worktree master form SHA256 at intake:
`0821e9ea94225ff6332a9ce55ac32eb518277eaf0dc50cd2ba3c1f36be0aa131`.
- Isolated worktree: `/private/tmp/kk-715-stack01-audit`, branch `audit/715-stack01-review`.
- Code-bearing [PR #831](#): `release/715-stack-01-evm` at `66ca726662822a9ff102bcd1354cc6a39d5a203b` against immediate predecessor `release/715-stack-00b-release-foundation` at `4c56e19e3070772a28b2890a4ec862ecf77d5429` when claimed.
- The adjacent diff is 80 paths, 3,017 added and 663 deleted lines, from `git diff --numstat 4c56e19e3070772a28b2890a4ec862ecf77d5429..66ca726662822a9ff102bcd1354cc6a39d5a203b`. It changes firmware, tests, CI, build inputs and documentation; the PR is code-bearing. The one commit in the range is `66ca72666`.
- Stack 00b is separately claimed and has no completed checkpoint in the canonical ledger. Stack 01 cannot be called accepted on predecessor ancestry until that claim completes and its head is reconciled.
- Audit remediation commits: `cbc725dc6` restores the emulator animation; `692543774` adds its board regression; `199e491e9` narrows auto-lock activity to validated workflow continuation responses and replaces direct helper tests with transport tests. `5e0151438` and `4f3daef30` are report intake edits. The containing final report head is recorded in the PR body.

Frozen dependency pins and original line inventory

The predecessor and original Stack 01 head have identical top-level gitlinks:

Dependency	Exact commit
<code>deps/crypto/trezor-firmware</code>	<code>8a392f70a5d5575ece3dfb35f115d4a4b27f497c</code>
<code>deps/device-protocol</code>	<code>dd9c85dc747cf965fb0e7bf49615dc9e7568ee65</code>
<code>deps/python-keepkey</code>	<code>49d537ce953b7524599bfc87f56b7a50a51a13a1</code>
<code>deps/qrenc/QR-Code-generator</code>	<code>6dfbfadad5d9303ed190d1c3cb7bec34b565b6ce8</code>
<code>deps/sca-hardening/SecAESSTM32</code>	<code>71d356a1141624994cf613bd2d2583892e8e6d5a</code>

The Python host's nested device-protocol is at dd9c85dc and ethereum-lists at 89a64f717e1690bb31adb3e4c38e23640357333c. code-signing-keys is at a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec. No gitlink changes occur in this block; the .gitmodules line is whitespace only. The untouched nested crypto test/vendor gitlinks were not initialized locally and are not inputs to the cited native build.

The original unit has one commit, 66ca726662822a9ff102bcd1354cc6a39d5a203b. Its complete per-path counts below come from the immutable command `git show --format= --numstat --no-renames 66ca726662822a9ff102bcd1354cc6a39d5a203b`. The counts include implementation, tests, workflows, scripts, build files and documentation; they are the original adjacent diff, not a sum of later audit remediation commits. The changed-path descriptions are grouped here to keep the per-path table readable:

- **Build, protocol and CI:** CMake now includes Zcash privacy in the regular image and keeps Bitcoin only reduced, links the needed Pallas and metadata objects, and checks native XML inputs. CI retries external pulls and caches cppcheck and images, while its final gate still depends on static analysis. The release instructions now match each binary against its own manifest. Full and Bitcoin-only hosted jobs passed at the original head.
- **Dice and setup:** The reset ceremony now stages settings until a seed is committed, offers two explicit dice modes with on-device consent, shows the device draw for offline verification in mixed mode, rejects biased rolls, and pages all backup words. The verifier reproduces the firmware tags and BIP-39 conversion. Host dice capture and unit cases check cancellation, derivation, settings and output. Physical display and power interruption remain release checks.
- **Session and secret lifetime:** Transport, tiny acknowledgements, PIN and passphrase buffers, RNG output, storage reads and CRC, recovery word count, and signing context receive targeted bounds, cleanup and fail-closed fixes. The local S01-01/S01-02 patches address the auto-lock policy and emulator host-wait animation found during this review.
- **Chain behavior:** Wanchain transfer formatting uses the current request; THOR/MAYA deposit memo narration, MAYA decimals, Ripple response lifetime and unsupported memo rejection, Bitcoin input comparison, and Zcash build coverage have corresponding native cases. Message-map entries were compared as parsed tuples: all 140 base entries equal all 140 head entries in order; its 411 changed lines are formatting and equivalent conditional spelling.
- **Prior audit notes:** The added historical P-unit notes and previous combined-candidate documents explain source provenance; they do not turn their receipt-only review PRs into firmware-code coverage. This block's code-bearing adjacent diff supplies the current review surface.

Original commit path counts

Added	Deleted	Path
65	15	.github/workflows/ci.yml
24	2	.github/workflows/release.yml
1	1	.gitmodules
16	4	CMakeLists.txt
1	0	deps/crypto/CMakeLists.txt
141	39	docs/DiceEntropy.md
154	0	docs/dice-vs-coldcard.md
78	0	docs/firmware/reviews/7.15.0-review-round2-remediation.md
83	31	docs/release/7.14.3-COMBINED-CANDIDATE.md
12	0	docs/release/audit-units/P01-004-report-inputs.md
16	0	docs/release/audit-units/P01-008-compose-variant.md
27	0	docs/release/audit-units/P02-001-tiny-lifetime.md
14	0	docs/release/audit-units/P02-003-packet-lifetime.md
18	0	docs/release/audit-units/P06-002-ripple-display-response.md
15	0	docs/release/audit-units/P06-003-ripple-display-test.md
21	0	docs/release/audit-units/P06-005-ripple-unsupported-memo.md
12	0	docs/release/audit-units/P06-006-ripple-memo-rejection-test.md
87	0	docs/release/audit-units/full-diff-audit-20260912.md
32	0	docs/release/audit-units/scope-repair.md
12	1	docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
3	0	include/keepkey/board/layout.h
3	0	include/keepkey/board/messages.h
26	3	include/keepkey/firmware/dice_input.h
7	10	include/keepkey/firmware/ethereum_contracts/thortx.h
1	0	include/keepkey/firmware/fsm.h
2	0	include/keepkey/firmware/home_sm.h
1	0	include/keepkey/firmware/mayachain.h
14	12	include/keepkey/firmware/reset.h
8	2	include/keepkey/firmware/ripple.h
1	1	include/keepkey/firmware/tiny-json.h
2	1	include/keepkey/firmware/txin_check.h
8	1	lib/board/confirm_sm.c
13	10	lib/board/keepkey_board.c
4	2	lib/board/keepkey_flash.c
4	3	lib/board/layout.c
38	10	lib/board/messages.c
2	5	lib/board/udp.c
15	3	lib/board/usb.c
6	0	lib/firmware/CMakeLists.txt
131	70	lib/firmware/dice_input.c
11	0	lib/firmware/ethereum.c

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3      2 lib/firmware/ethereum_contracts/thortx.c
52    16 lib/firmware/fsm.c
23     7 lib/firmware/fsm_msg_common.h
12     4 lib/firmware/fsm_msg_ripple.h
37     1 lib/firmware/home_sm.c
16     1 lib/firmware/mayachain.c
245   166 lib/firmware/messagemap.def
1      0 lib/firmware/passphrase_sm.c
20     8 lib/firmware/pin_sm.c
22     2 lib/firmware/recovery_cipher.c
232   130 lib/firmware/reset.c
1      1 lib/firmware/ripple.c
15     6 lib/firmware/signing.c
32    11 lib/firmware/storage.c
2      2 lib/firmware/transaction.c
6      4 lib/firmware/txin_check.c
11     0 lib/rand/rng_health.c
19     9 scripts/emulator/capture-dice-flow.py
1      0 scripts/emulator/docker-compose.bitcoin-only.yml
21     2 scripts/generate-test-report.py
1      1 tools/firmware/keepkey.ld
251   0   tools/verify_dice_seed.py
46     0 unittests/board/board.cpp
1      0 unittests/firmware/CMakeLists.txt
33     0 unittests/firmware/confirm_test_utils.cpp
78    26 unittests/firmware/dice.cpp
242   0   unittests/firmware/fsm.cpp
19     0 unittests/firmware/mayachain.cpp
46     0 unittests/firmware/recovery.cpp
20     0 unittests/firmware/ripple.cpp
25     0 unittests/firmware/rng_health.cpp
15     9 unittests/firmware/setup_ceremony.cpp
21    22 unittests/firmware/signed_metadata.cpp
48     0 unittests/firmware/signing.cpp
33     1 unittests/firmware/storage.cpp
26     0 unittests/firmware/storage_passphrase.cpp
102   0   unittests/firmware/transaction.cpp
95     0 unittests/firmware/usb_rx.cpp
15     6 unittests/firmware/zcash.cpp

```

Evidence checked

- `git diff --check` on the frozen range passed.
- Exact-head nonpublishing [CI run 35685552039](#) succeeded: format, secret scan, static analysis, submodule check, full and Bitcoin-only ARM builds and SRAM gates, emulator builds, unit and Python integration suites, dylib tests, report generation, evidence and CI gates. `publish-emulator` was skipped as intended. A concurrent run 35685552437 at the same SHA was cancelled; it is not the success receipt.
- Original run artifacts include full and Bitcoin-only ARM bundles 10677175575 and 10676522831, test report 10677076880, full and Bitcoin-only native results 10676681571 and 10676557872, Python results 10676921425 and 10676557869, and OLED captures 10676691543 and 10676338173. The two emulator-image artifacts have expired; all these artifacts refer to the original PR head, not the local remediation.
- Source inspection covered `reset/dice` derivation and pager, transport rejection and tiny replies, USB buffer cleanup, FSM session policy, storage CRC, signing input comparison, Ripple, MAYAChain, the message map and CI wiring. The complete original changed-path ledger and dependency pin reconciliation are above. Physical display review remains a release gate.
- The local emulator fix and its focused `BodyFits.EmulatorPollAdvancesHostWaitProgress` regression passed in the SOP Docker `firmware-unit` compose run on 2026-09-23: 327 firmware, 20 board and 18 crypto tests passed. The first local build attempt stopped before compilation because the nested pinned `ethereum-lists` submodule was absent; after initializing it at 89a64f71, the same command completed with exit code 0. The Bitcoin-only compose run also passed: 114 firmware, 20 board and 18 crypto tests. The local fix commits are cbc725dc6 (pump) and 692543774 (regression); these are not in PR #831's green hosted run.

Findings and policy disposition

S01-01 — unrelated host traffic extends an active session's auto-lock

Severity: P2 policy/security review, fixed locally after owner decision. Changed lines: `lib/firmware/fsm.c:328–330` and `lib/firmware/home_sm.c:193–197` at the PR head. `fsm_usb_rx()` calls `note_host_activity()` before parsing or dispatching every frame. While `home_state == AWAY_FROM_HOME`, that clears the idle timer regardless of message type, validity or whether it advances the signing/setup workflow. A host that periodically sends `GetFeatures` (or malformed frames) can keep an otherwise

stalled signing session from reaching the configured auto-lock delay. The new tests in `unittests/firmware/fsm.cpp` call `note_host_activity()` directly and prove only that any call extends the timer; they do not exercise unrelated frames through USB or set a policy for them. The same concern was recorded in the 2026-09-13 re-audit handoff for the candidate line. On 2026-09-23 the owner specified that only active workflow progress should extend the timer. The local patch removes the pre-dispatch reset and calls `note_host_activity()` after a continuation request is encoded and sent. It names the supported streamed request types; Features, failures, and other unrelated responses are excluded. The new regression sends a valid `GetFeatures` packet through the parser during a Bitcoin signing wait and proves auto-lock still fires, while a sent `TxRequest` extends the wait. These tests passed in a full local Docker native run: 328 firmware, 20 board, 18 crypto tests. A compose up first failed before tests because local port 5000 was occupied; `docker compose run --no-deps firmware-unit` on the same built image completed with exit code 0. The original exact-head hosted CI predates this patch. The Bitcoin-only image built separately with `docker-compose.bitcoin-only.yml`, then its native run also passed: 115 firmware, 20 board, 18 crypto tests.

S01-02 — emulator progress animation stopped while polling the host

Severity: P3, fixed locally in this audit branch. The PR replaced the tail of `lib/board/udp.c:usbPoll()` with a buffer scrub and removed its call to `layout_animate_poll()`. The hardware USB poll still calls the animation pump; without the emulator call, a queued Zcash progress animation freezes whenever the emulator is waiting for host proof generation. The local patch restores the animation call after the buffer scrub. A new board test arms the real progress trickle, calls `emulator usbPoll()`, and requires a display refresh; it passed in the local Docker run above. Physical Zcash signing is still a release gate.

Current-source reconciliation

Surface	Current source and disposition	Evidence or limit
Session auto-lock	S01-01 fixed locally: only sent continuation requests reset the timer while away from home.	Full and Bitcoin-only native regression through <code>GetFeatures</code> RX and <code>TxRequest</code> TX passed. Hosted final-head checks pending.
Emulator host-wait progress	S01-02 fixed locally: <code>usbPoll()</code> scrubs its buffer and pumps <code>layout_animate_poll()</code> .	Board regression passed in both variants. Physical Zcash proof wait remains unverified.
Setup/dice derivation	<code>reset_init()</code> stages settings, commits only after backup review, and derives dice modes from recorded inputs. Firmware and offline verifier tags and SHA order match.	Native setup/dice cases and original hosted Python integration passed; physical OLED paging and power-cycle remain release gates.
Transport/secret lifetime	Tiny rejection unwinds once, clears buffers; raw RX and response arena lifetimes are bounded; new RNG draw recheck and V17 CRC alignment are in source.	Native unit coverage and original hosted ARM/SRAM pass. The test's emulator CRC matches the STM32 word recurrence.
Bitcoin signing	The current input digest is finalized from a copy; a refused output does not replace the prior comparison key; multisig sighash scratch has one extra byte.	Native signing/transaction cases passed. Physical signing remains a release gate.
Chain UX	Wanchain ticker uses current transaction type; THOR/MAYA raw memo and MAYA.CACAO scaling are present; Ripple rejects unsupported nonempty memos and copies the confirmed address after shared response arena use.	Native chain cases and original hosted integration passed. Router addresses remain externally maintained trust pins.
Storage power interruption	Inherited erase-before-replacement risk remains in this candidate.	Not fixed by Stack 01; signed-device power-cut and upgrade checks are still required before release.

No dependency pointer changed in the adjacent Stack 01 diff. The firmware source is on the code-bearing PR, while prior P02/P06 receipts on documentation PRs are historical evidence only. No physical device or signed image was used for this unit review.

Code-bearing review coverage

Claimed behavior	PR #831 code/test paths in the adjacent diff
Auto-lock counts workflow progress (S01-01)	<code>lib/firmware/fsm.c</code> , <code>lib/firmware/home_sm.c</code> , <code>lib/board/usb.c</code> , <code>include/keepkey/board/messages.h</code> , <code>unittests/firmware/fsm.cpp</code>
Emulator host-wait animation (S01-02)	<code>lib/board/udp.c</code> , <code>unittests/board/board.cpp</code>

Claimed behavior	PR #831 code/test paths in the adjacent diff
Dice and staged setup	lib/firmware/reset.c, lib/firmware/dice_input.c, tools/verify_dice_seed.py, unittests/firmware/dice.cpp, unittests/firmware/setup_ceremony.cpp
Storage and transport lifetime	lib/firmware/storage.c, lib/board/messages.c, lib/board/usb.c, lib/firmware/recovery_cipher.c, unittests/firmware/storage.cpp, unittests/firmware/usb_rx.cpp
Signing and chain display	lib/firmware/signing.c, lib/firmware/transaction.c, lib/firmware/txin_check.c, lib/firmware/ethereum.c, lib/firmware/fsm_msg_ripple.h, lib/firmware/mayachain.c, corresponding native tests
Variant and evidence gates	CMakeLists.txt, lib/firmware/CMakeLists.txt, .github/workflows/ci.yml, scripts/generate-test-report.py, tools/firmware/keepkey.ld

The late reviewer must be given the final base-to-head diff for these paths; a review of this report alone would assess only the receipt.

Verification matrix and limits

Check	Exact source	Result	Limit
Original full CI	66ca726662822a9ff102bcd1354cc6a39d5a203b, run 35685552039	Green required graph: ARM/SRAM, emulator, native, Python, dylib, static, lint, report and CI gate.	Predates audit fixes; emulator publication skipped.
Original concurrent CI	same SHA, run 35685552437	Cancelled.	Not a success receipt.
Full local native	audit code through 199e491e9	328 firmware, 20 board, 18 crypto passed.	Docker AMD64 under ARM host; native emulator tests, no physical device.
Bitcoin-only local native	audit code through 199e491e9	115 firmware, 20 board, 18 crypto passed.	Same host limit.
Full local host/emulator	audit code through 199e491e9	Docker Python suite exited 0: 568 passed, 262 skipped; OLED-required capture pass ran first.	Uses the same missing-capability list as original CI. Skip set includes still-unshipped release capabilities; no physical OLED.
Local report input gate	audit code through 199e491e9	require_native_junit() accepted the three extracted nonempty, parseable JUnit suites: firmware.xml, board.xml, crypto.xml.	This exercises the changed native-input gate, not the full report assembly, which also needs CI ARM and dylib artifacts.
Original diff whitespace	4c56e19e..66ca7266	git diff --check passed.	Will rerun on final range.
Message map parity	original predecessor and head	All 140 parsed entries equal in order.	Does not prove handler semantics by itself.

The original CI ARM bundles still bear the staging filename v7.14.3; that filename does not establish a 7.15 signed release artifact. The local code patch has no new dependency or workflow changes. A green CI run on the pushed final code-bearing head is required before a unit checkpoint.

Review PR inventory and local preflight

The first frozen report commit is a4c4d96f2 (report source and PDF). The immutable inventory reproduction range for the complete code-bearing PR at that point is `git diff --numstat 4c56e19e3070772a28b2890a4ec862ecf77d5429..a4c4d96f2`. It contains 82 paths, 3,397 added and 662 deleted text lines, plus one binary PDF with `-/-` counts. Every path in that range is listed below. The subsequent correction that embeds this table changes only this report source and its PDF; the final containing commit and its exact final numeric diff are recorded in the PR body. No code, test, pin or workflow file changes after a4c4d96f2 are carried forward without a new preflight.

Added	Deleted	Path
65	15	.github/workflows/ci.yml
24	2	.github/workflows/release.yml
1	1	.gitmodules
16	4	CMakeLists.txt

```

1      0 deps/crypto/CMakeLists.txt
141    39 docs/DiceEntropy.md
154    0 docs/dice-vs-coldcard.md
78     0 docs/firmware/reviews/7.15.0-review-round2-remediation.md
83    31 docs/release/7.14.3-COMBINED-CANDIDATE.md
305    0 docs/release/audit-units/715-01-review-20260923.md
-      - docs/release/audit-units/715-01-review-20260923.pdf
12     0 docs/release/audit-units/P01-004-report-inputs.md
16     0 docs/release/audit-units/P01-008-compose-variant.md
27     0 docs/release/audit-units/P02-001-tiny-lifetime.md
14     0 docs/release/audit-units/P02-003-packet-lifetime.md
18     0 docs/release/audit-units/P06-002-ripple-display-response.md
15     0 docs/release/audit-units/P06-003-ripple-display-test.md
21     0 docs/release/audit-units/P06-005-ripple-unsupported-memo.md
12     0 docs/release/audit-units/P06-006-ripple-memo-rejection-test.md
87     0 docs/release/audit-units/full-diff-audit-20260912.md
32     0 docs/release/audit-units/scope-repair.md
12     1 docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
3      0 include/keepkey/board/layout.h
7      0 include/keepkey/board/messages.h
26     3 include/keepkey/firmware/dice_input.h
7      10 include/keepkey/firmware/ethereum_contracts/thortx.h
1      0 include/keepkey/firmware/fsm.h
2      0 include/keepkey/firmware/home_sm.h
1      0 include/keepkey/firmware/mayachain.h
14     12 include/keepkey/firmware/reset.h
8      2 include/keepkey/firmware/ripple.h
1      1 include/keepkey/firmware/tiny-json.h
2      1 include/keepkey/firmware/txin_check.h
8      1 lib/board/confirm_sm.c
13     10 lib/board/keepkey_board.c
4      2 lib/board/keepkey_flash.c
4      3 lib/board/layout.c
38     10 lib/board/messages.c
3      4 lib/board/udp.c
22     3 lib/board/usb.c
6      0 lib/firmware/CMakeLists.txt
131    70 lib/firmware/dice_input.c
11     0 lib/firmware/ethereum.c
3      2 lib/firmware/ethereum_contracts/thortx.c
69     16 lib/firmware/fsm.c
23     7 lib/firmware/fsm_msg_common.h
12     4 lib/firmware/fsm_msg_ripple.h
37     1 lib/firmware/home_sm.c
16     1 lib/firmware/mayachain.c
245    166 lib/firmware/message_map.def
1      0 lib/firmware/passphrase_sm.c
20     8 lib/firmware/pin_sm.c
22     2 lib/firmware/recovery_cipher.c
232    130 lib/firmware/reset.c
1      1 lib/firmware/ripple.c
15     6 lib/firmware/signing.c
32     11 lib/firmware/storage.c
2      2 lib/firmware/transaction.c
6      4 lib/firmware/txin_check.c
11     0 lib/rand/rng_health.c
19     9 scripts/emulator/capture-dice-flow.py
1      0 scripts/emulator/docker-compose.bitcoin-only.yml
21     2 scripts/generate-test-report.py
1      1 tools/firmware/keepkey.ld
251    0 tools/verify_dice_seed.py
61     0 unittests/board/board.cpp
1      0 unittests/firmware/CMakeLists.txt
33     0 unittests/firmware/confirm_test_utils.cpp
78     26 unittests/firmware/dice.cpp
273    0 unittests/firmware/fsm.cpp
19     0 unittests/firmware/mayachain.cpp
46     0 unittests/firmware/recovery.cpp
20     0 unittests/firmware/ripple.cpp
25     0 unittests/firmware/rng_health.cpp
15     9 unittests/firmware/setup_ceremony.cpp
21     22 unittests/firmware/signed_metadata.cpp
48     0 unittests/firmware/signing.cpp
33     1 unittests/firmware/storage.cpp
26     0 unittests/firmware/storage_passphrase.cpp
102    0 unittests/firmware/transaction.cpp
95     0 unittests/firmware/usb_rx.cpp
15     6 unittests/firmware/zcash.cpp

```

The code-bearing gate scripts/release/check-code-review-diff.sh 4c56e19e3070772a28b2890a4ec862ecf77d5429 a4c4d96f2 passed and printed the source, test, build and workflow paths mapped above. git diff --check passed on the same immutable range. The live PR base/head at preflight were 4c56e19e3070772a28b2890a4ec862ecf77d5429 and

66ca726662822a9ff102bcd1354cc6a39d5a203b, respectively; the remote head must be rechecked after push. The PDF was rendered from this Markdown source with WeasyPrint, text extracted, and all six pages visually inspected for legibility. The final PDF is regenerated with this inventory correction.

Checkpoint status

- Internal source and interaction review: complete on the local candidate; S01-01 and S01-02 are fixed and exercised in both variants. The inherited storage power-interruption risk is unchanged and remains a release blocker.
- Exact-head hosted CI: pending until the report and code head are pushed.
- Stack 00b: separately claimed, not yet accepted; reconcile its final head and PR #831's base before accepting Stack 01.
- Late code-bearing Copilot checkpoint: not requested yet. This report and its PDF must be frozen and preflighted first; review must cover PR #831's code paths on its current base and head.
- Assembled 7.15 product and physical release: pending signed-device upgrade, power-interruption, signing, Zcash wait and OLED checks. No production promotion or publication is implied by this unit audit.